

Student Name: _____

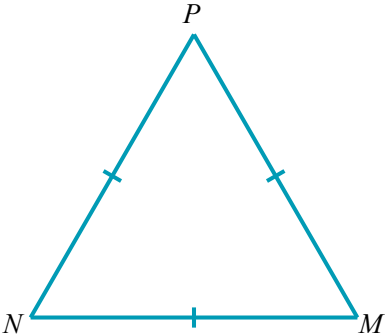
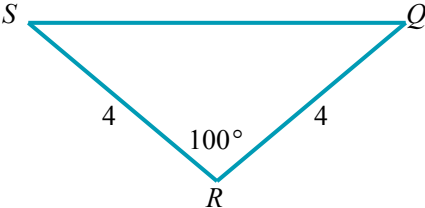
Class Name: **Geo - Graybeal - 23-24 - C**

Number of Questions: **14**

Instructor Name: **Graybeal, Michael**

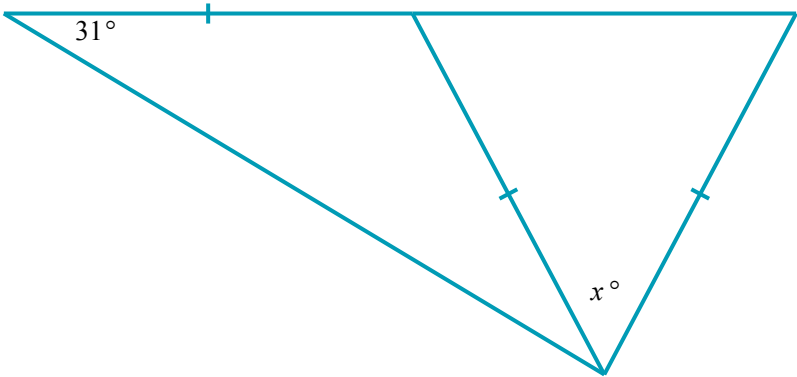
Question 1 of 14

For each part below, use the figure to fill in the blank.
If necessary, you may learn what the markings on a figure indicate.

(a) Find $m\angle M$.  $m\angle M = \boxed{}^\circ$	(b) Find $m\angle Q$.  $m\angle Q = \boxed{}^\circ$
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Question 2 of 14

Find the value of x .
If necessary, you may learn what the markings on a figure indicate.

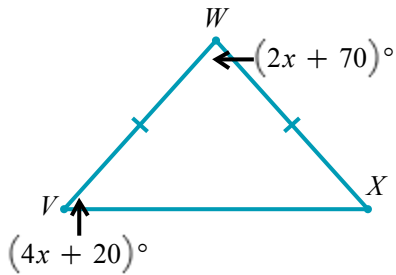


Question 3 of 14

Suppose that $\triangle VWX$ is isosceles with base $\overline{VX}$.

Suppose also that $m\angle V = (4x + 20)^\circ$ and $m\angle W = (2x + 70)^\circ$.

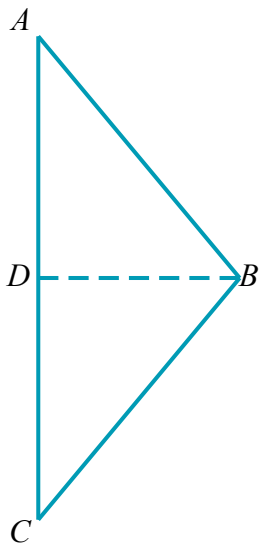
Find the degree measure of each angle in the triangle.



Question 4 of 14

Use the given information to complete the proof of the following theorem.

The angles opposite the two congruent sides of an isosceles triangle are congruent.



Given: $\overline{BA} \cong \overline{BC}$

$\overline{BD}$ bisects $\angle ABC$

Prove: $\angle A \cong \angle C$

Question 5 of 14

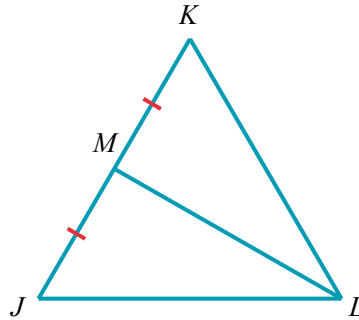
Classify the segment in each part below.

Only use the information given in the figure.

Do not assume measures are equal unless the markings indicate they are.

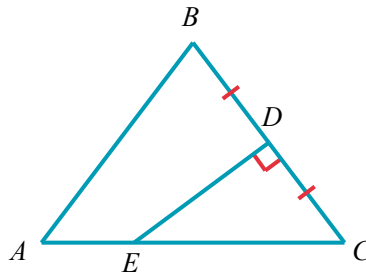
(a) Select *all* that describe $\overline{LM}$.

- ☐ Perpendicular bisector of $\overline{JK}$
- ☐ Angle bisector of $\angle L$
- ☐ Median of $\triangle JKL$
- ☐ Altitude of $\triangle JKL$
- ☐ None of the above



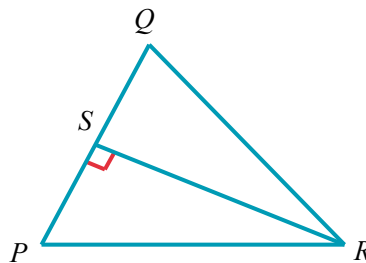
(b) Select *all* that describe $\overline{DE}$.

- ☐ Perpendicular bisector of $\overline{BC}$
- ☐ Angle bisector of $\angle A$
- ☐ Median of $\triangle ABC$
- ☐ Altitude of $\triangle ABC$
- ☐ None of the above



(c) Select *all* that describe $\overline{RS}$.

- ☐ Perpendicular bisector of $\overline{PQ}$
- ☐ Angle bisector of $\angle R$
- ☐ Median of $\triangle PQR$
- ☐ Altitude of $\triangle PQR$
- ☐ None of the above



Question 6 of 14

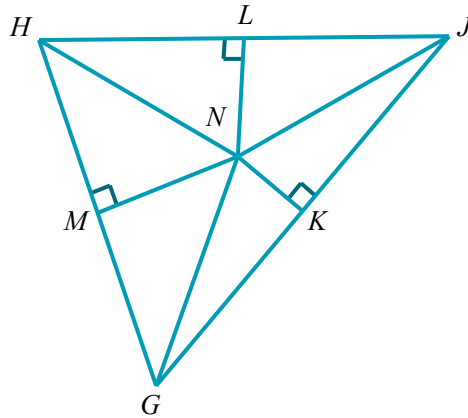
Consider $\triangle GHJ$ in the figure below.

The perpendicular bisectors of its sides are $\overline{KN}$, $\overline{LN}$, and $\overline{MN}$. They meet at a single point N . (In other words, N is the circumcenter of $\triangle GHJ$.)

Suppose $LN = 48$, $GK = 88$, and $GN = 102$.

Find HN , LJ , and GJ .

Note that the figure is not drawn to scale.



$$HN = \underline{\hspace{2cm}}$$

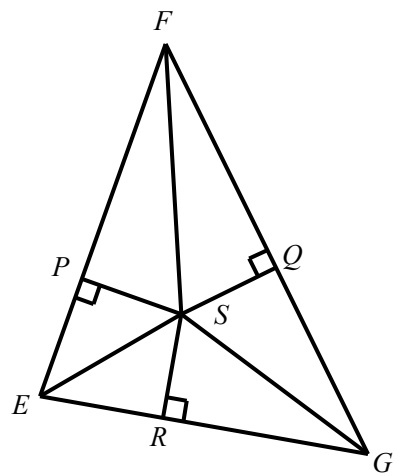
$$LJ = \underline{\hspace{2cm}}$$

$$GJ = \underline{\hspace{2cm}}$$

Question 7 of 14

The angle bisectors of $\triangle EFG$ are $\overline{ES}$, $\overline{FS}$, and $\overline{GS}$. They meet at a single point S .
(In other words, S is the incenter of $\triangle EFG$.)

Suppose $PS = 16$, $ES = 21$, $m\angle PER = 92^\circ$, and $m\angle RGS = 18^\circ$.
Find the following measures.
Note that the figure is not drawn to scale.



$m\angle RGQ = \underline{\hspace{1cm}}^\circ$
 $m\angle QFS = \underline{\hspace{1cm}}^\circ$
 $QS = \underline{\hspace{1cm}}$

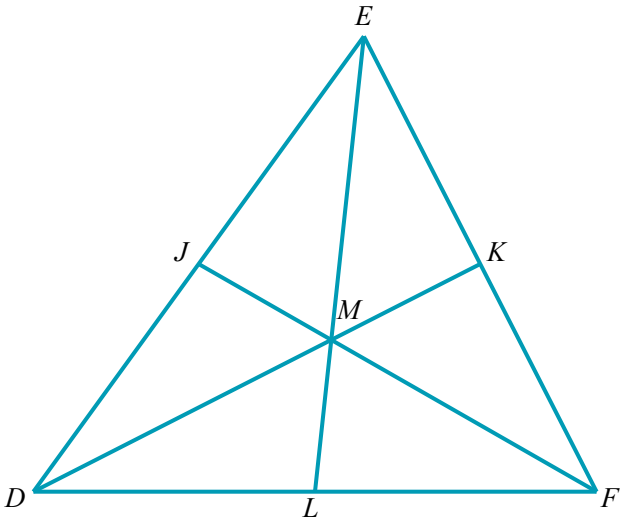
Question 8 of 14

The medians of $\triangle DEF$ are $\overline{DK}$, $\overline{EL}$, and $\overline{FJ}$. They meet at a single point M . (In other words, M is the centroid of $\triangle DEF$.)

Suppose $EM = 12$, $FJ = 60$, and $MK = 11$.

Find the following lengths.

Note that the figure is not drawn to scale.



$MJ = \underline{\hspace{2cm}}$

$DM = \underline{\hspace{2cm}}$

$EL = \underline{\hspace{2cm}}$

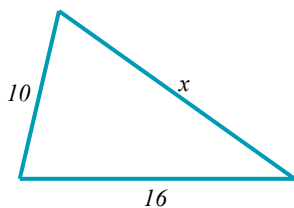
Question 9 of 14

For each set of three lengths, determine if they can be the side lengths of a triangle.

Lengths	Can be side lengths of a triangle	Cannot be side lengths of a triangle
4.8, 11.4, 5.9	☐	☐
8, 16, 9	☐	☐
11, 3, 11	☐	☐
3, 16, 8	☐	☐

Question 10 of 14

The side lengths of a triangle are 10, 16, and x units.



(The figure above gives only one possible configuration for the triangle.)

Determine the possible values of x .

Write your answer as an inequality.

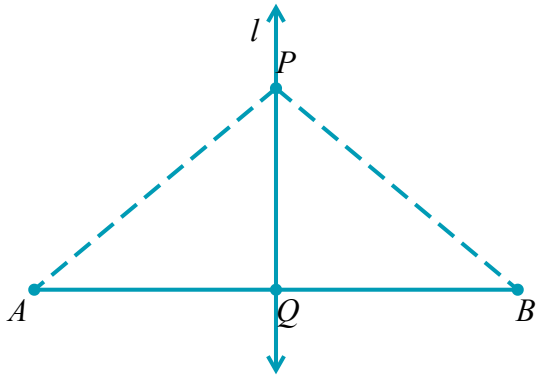
Use x only once in your inequality.

Question 11 of 14

Use the given information to prove the following theorem.

If a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment.

We let P be any point on line l , but different from point Q .



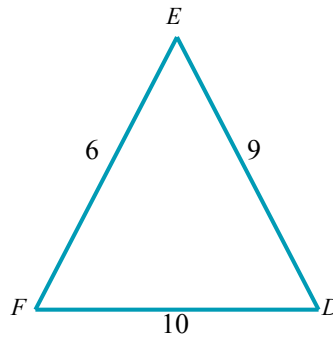
Given: $\overline{PQ}$ is the $\perp$ bisector of $\overline{AB}$

Prove: $AP = BP$

Question 12 of 14

Order the angle measures $m\angle D$, $m\angle E$, and $m\angle F$ from least to greatest.

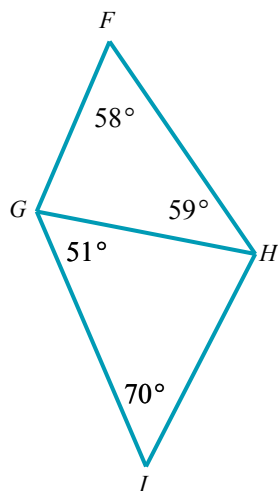
(Note that the figure is not drawn to scale.)



Question 13 of 14

Consider the following figure.

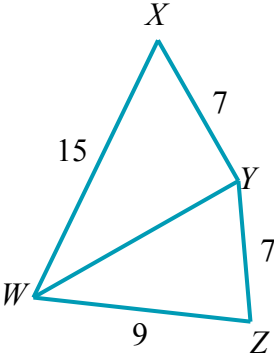
(Note that the figure is not drawn to scale.)



Order the side lengths FG , GI , FH , GH , and IH from least to greatest.

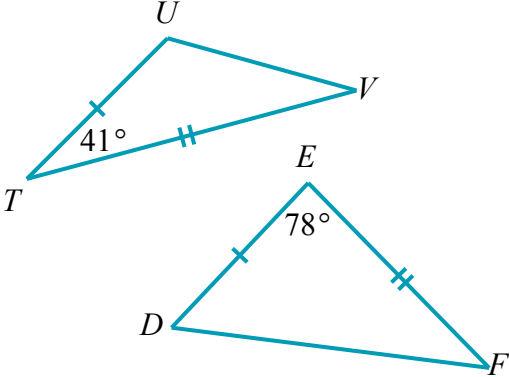
Question 14 of 14

Compare the given measures by choosing $<$, $>$, or $=$.
If there is not enough information, select "Cannot be determined".
If necessary, you may learn what the markings on a figure indicate.



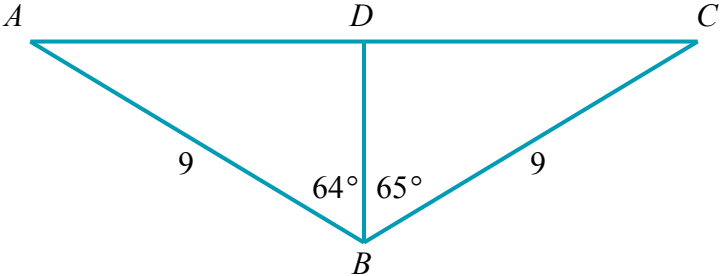
$\bigcirc m\angle WYX (</>/=) m\angle WYZ$

$\bigcirc$ Cannot be determined



$\bigcirc UV (</>/=) DF$

$\bigcirc$ Cannot be determined



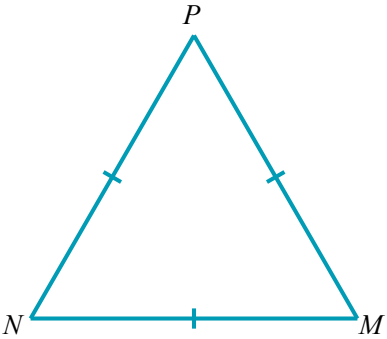
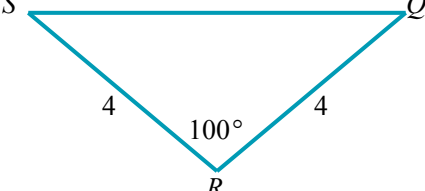
$\bigcirc AD (</>/=) CD$

$\bigcirc$ Cannot be determined

Class Name: **Geo - Graybeal - 23-24 - C**

Number of Questions: **14**

Question 1 of 14

(a) Find $m\angle M$.  $m\angle M = 60^\circ$	(b) Find $m\angle Q$.  $m\angle Q = 40^\circ$
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Question 2 of 14

$x = 56$

Question 3 of 14

$m\angle V = 48^\circ$
 $m\angle W = 84^\circ$
 $m\angle X = 48^\circ$

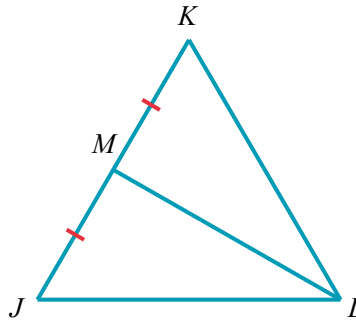
Question 4 of 14

Statement	Reason
1 $\overline{BA} \cong \overline{BC}$	Given
2 $\overline{BD}$ bisects $\angle ABC$	Given
3 $\angle ABD \cong \angle CBD$	Definition of Angle Bisector
4 $\overline{BD} \cong \overline{BD}$	Reflexive Property
5 $\triangle BAD \cong \triangle BCD$	SAS Congruence Property
6 $\angle A \cong \angle C$	CPCTC Property

Question 5 of 14

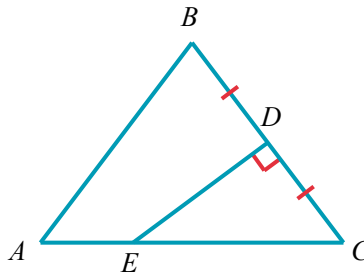
(a) Select *all* that describe $\overline{LM}$.

- ☐ Perpendicular bisector of $\overline{JK}$
- ☐ Angle bisector of $\angle L$
- ☒ Median of $\triangle JKL$
- ☐ Altitude of $\triangle JKL$
- ☐ None of the above



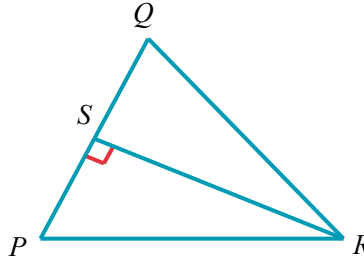
(b) Select *all* that describe $\overline{DE}$.

- ☒ Perpendicular bisector of $\overline{BC}$
- ☐ Angle bisector of $\angle A$
- ☐ Median of $\triangle ABC$
- ☐ Altitude of $\triangle ABC$
- ☐ None of the above



(c) Select *all* that describe $\overline{RS}$.

- ☐ Perpendicular bisector of $\overline{PQ}$
- ☐ Angle bisector of $\angle R$
- ☐ Median of $\triangle PQR$
- ☒ Altitude of $\triangle PQR$
- ☐ None of the above



Question 6 of 14

$$HN = 102$$

$$LJ = 90$$

$$GJ = 176$$

Question 7 of 14

$m\angle RGQ=36^\circ$

$m\angle QFS=26^\circ$

$QS=16$

Question 8 of 14

$MJ=20$

$DM=22$

$EL=18$

Question 9 of 14

Lengths	Can be side lengths of a triangle	Cannot be side lengths of a triangle
4.8, 11.4, 5.9	☐	☒
8, 16, 9	☒	☐
11, 3, 11	☒	☐
3, 16, 8	☐	☒

Question 10 of 14

$6<x<26$

Question 11 of 14

For this proof, you should be familiar with the following definitions and properties.

- Definition of segment bisector
- Definition of perpendicular bisector
- Definition of perpendicular
- Definition of equidistant
- Definition of right angle
- Reflexive property
- Triangle congruence properties
- CPCTC Property
- Definition of congruent

We are proving the following theorem.

If a point is on the perpendicular bisector of a segment, then it is equidistant from the endpoints of the segment.

Understanding the theorem:

Refer to Figure 1. $\overline{PQ}$ is the $\perp$ bisector of $\overline{AB}$.

Point P represents *any* point on line l other than Q . The theorem states that P must be equidistant from the endpoints A and B .

So, to prove the theorem, we must prove that $AP = BP$.

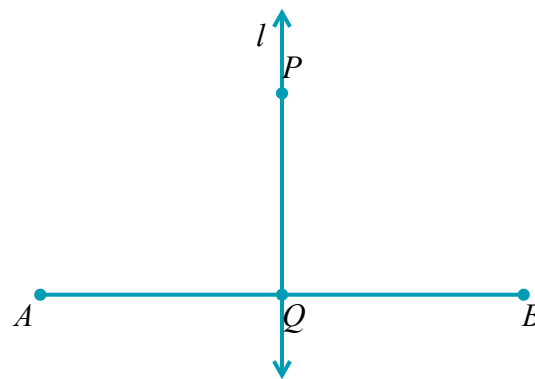


Figure 1

Proving the theorem:

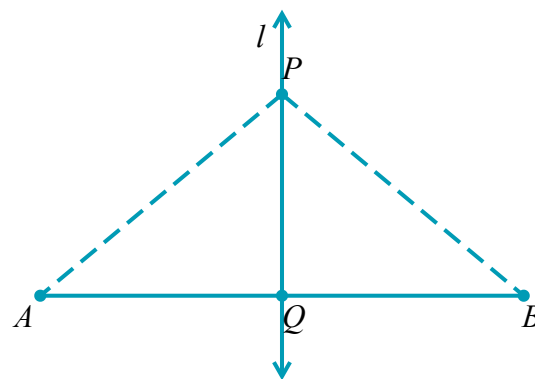


Figure 2

We start by drawing $\overline{PA}$ and $\overline{PB}$. Notice that this creates $\triangle AQP$ and $\triangle BQP$.

We can prove $AP = BP$ as follows.

1. We are given that $\overline{PQ}$ is the $\perp$ bisector of $\overline{AB}$.
2. By the definition of a perpendicular bisector, statement 1 implies that $\overline{AQ} \cong \overline{BQ}$. This gives us a pair of congruent sides.
3. By the definition of a perpendicular bisector, statement 1 implies that $\angle AQP$ is a right angle.
4. Again, by the definition of a perpendicular bisector, statement 1 implies that $\angle BQP$ is a right angle.
5. All right angles are congruent, therefore $\angle AQP \cong \angle BQP$. This gives us a pair of congruent angles.
6. By the reflexive property of congruence, $\overline{PQ}$ is congruent to $\overline{PQ}$. This gives us a pair of congruent sides since $\overline{PQ}$ is a part of both triangles.
7. Now we have $\triangle AQP \cong \triangle BQP$ by the SAS Congruence Property.
8. So, $\overline{AP} \cong \overline{BP}$ by the CPCTC Property.
9. Finally, $AP = BP$ by the Definition of Congruent Segments.

It is important to note the following:

In the proof, we do not specify how far P is from Q . (The figure gives only one possible example.) So, we showed that any point P on line l , other than the point Q , is equidistant from A and B . The point Q is the midpoint of segment $\overline{AB}$, so it is also equidistant from A and B . Thus, the theorem is true in general (for all points on line l).

Here is one possible answer.

Question 12 of 14

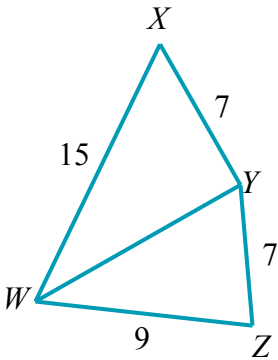
$$m\angle D < m\angle F < m\angle E$$

Question 13 of 14

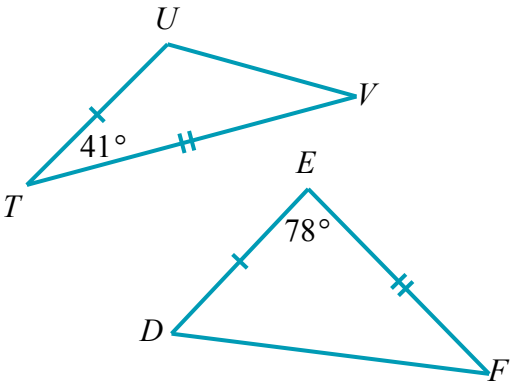
Order the side lengths FG , GI , FH , GH , and IH from least to greatest.

$$IH < GI < GH < FG < FH$$

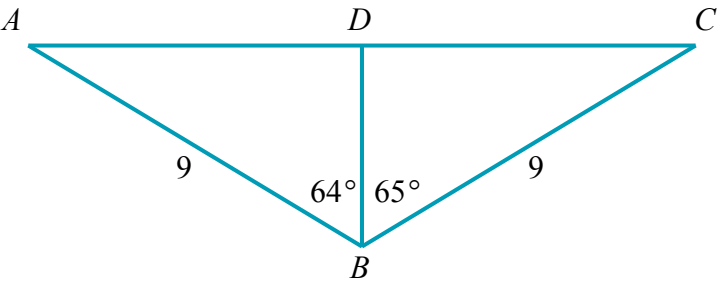
Question 14 of 14



- ☒ $m\angle WYX > m\angle WYZ$
- ☐ Cannot be determined



- ☒ $UV < DF$
- ☐ Cannot be determined



- ☒ $AD < CD$
- ☐ Cannot be determined